

Risk-Calibrated Test-Time Decision Making for Reliable Retrieval-Augmented Generation

Anonymous

Paper ID XXX

Abstract

Retrieval-Augmented Generation (RAG) improves factual grounding in Large Language Models (LLMs) but does not guarantee reliability: retrieved evidence may be irrelevant, contradictory, or misleading, leading to confident errors. Existing approaches treat uncertainty quantification and abstention separately, offering only binary answer-or-abstain decisions. We formalize the *RAG test-time risk control* problem with four actions—ANSWER, RETRIEVE MORE, VERIFY MORE, and ABSTAIN—under a limited compute budget, and propose **RC-TTD** (Risk-Calibrated Test-Time Decision Policy), a two-stage method combining conformal prediction with a retrieval-aware nonconformity score. Stage 1 uses answer probability ($1-p_{\max}$) for conformal coverage ($\geq 1-\alpha$), while Stage 2 leverages a multi-signal score—fusing answer uncertainty, retrieval relevance, evidence conflict, answer-evidence support, and verifier disagreement—to select corrective actions. We prove five theoretical results: (1) marginal score coverage, (2) a formal *score distribution compression* lemma with explicit assumptions showing why naive combined-score calibration fails, (3) a selective risk decomposition theorem, (4) a cost-optimality theorem over general single-action policies, and (5) a risk-coverage trade-off theorem showing strict risk improvement at matched coverage. We also formally characterize the *confident error* limitation. Experiments on SciQ, SciFact, and wrong-context stress tests with LLaMA-3.1-8B use *real end-to-end corrective actions* (re-retrieval with top-15, per-evidence NLI-weighted verification), showing RC-TTD maintains target coverage ($\sim 90\%$), achieves strictly lower selective risk than probability-only abstention at matched coverage, and provides provably better cost-risk trade-offs. Under wrong-context corruption, where LAC/APS become over-conservative (coverage drops to 0.81), RC-TTD maintains **$\sim 100\%$ coverage** with RETRIEVE MORE fix rates up to 36%.

Introduction

Retrieval-Augmented Generation (RAG) [Lewis et al.(2020), Gao et al.(2024)] has become the standard approach for grounding Large Language Model (LLM) outputs in external knowledge, reducing hallucinations [Ji et al.(2023), Shuster et al.(2021)] by incorporating retrieved evidence at inference time. However, RAG systems are not inherently reliable: the retrieval pipeline may return irrelevant documents, the evidence may be contradictory, or the model may be misled by plausible-but-incorrect

context. These failure modes produce *confident errors*—cases where the model generates a wrong answer with high confidence [Kadavath et al.(2022), Mallen et al.(2023)]—which are particularly dangerous in high-stakes domains such as healthcare, science, and finance. Compounding this, recent work [Schaeffer et al.(2024)] shows that such confident errors are not rare tail events but systematic failure modes under distribution shift.

Existing approaches to RAG reliability fall into three lines of work, each with limitations. First, **uncertainty quantification** benchmarks like URAG [Ye et al.(2025)] systematically evaluate RAG confidence calibration but do not prescribe what action to take when uncertainty is high; similarly, language-model self-knowledge studies [Kadavath et al.(2022), Lin, Hilton, and Evans(2022), Tian et al.(2023), Kuhn, Gal, and Farquhar(2023)] provide richer confidence signals but stop short of decision-making. Second, **conformal abstention** methods [Mohri et al.(2024), Quach et al.(2024), Sadinle, Lei, and Wasserman(2019), Romano et al.(2020)] provide distribution-free risk guarantees [Vovk, Gammerman, and Shafer(2005), Angelopoulos and Bates(2023)] but are limited to binary answer-or-abstain decisions, ignoring RAG-specific corrective actions such as retrieving additional evidence or verifying the answer against multiple sources. Third, **test-time compute scaling** research [Snell et al.(2024)Snell, Lee, Xu, and Kumar, Lightman et al.(2023)Lightman, Kosaraju, Burda et al.] studies how to allocate additional computation between solving and verifying, but does not address RAG-specific signals like retrieval relevance or evidence conflict, nor does it provide calibrated risk control.

The key insight of this paper is that RAG systems benefit from a *multi-action* decision framework that goes beyond binary abstention. When a RAG system is uncertain about an answer, the appropriate response depends on *why* it is uncertain: if retrieval is irrelevant, the system should retrieve more evidence; if evidence is contradictory, it should verify the answer; if the model is fundamentally unsure even with good evidence, it should abstain. This requires a risk-calibrated policy that maps the RAG system’s internal signals to one of four actions under a limited compute budget.

We formalize this as the *RAG test-time risk control* prob-

lem and propose **RC-TTD** (Risk-Calibrated Test-Time Decision Policy), a two-stage method that decouples conformal coverage guarantees from corrective action selection. Figure 1 illustrates the overall pipeline. Our main contributions are:

1. **Problem formulation:** We define the RAG test-time multi-action risk control problem with four actions (ANSWER, RETRIEVE MORE, VERIFY MORE, ABSTAIN) under a compute budget, incorporating RAG-specific signals.
2. **Retrieval-aware nonconformity score:** We design a multi-signal score fusing five uncertainty diagnostics and demonstrate through ablation that each signal contributes to risk prediction quality.
3. **Theoretical analysis with five results:** We prove (i) marginal score coverage (Theorem 1), (ii) a *score distribution compression lemma* (Lemma 1) with explicit assumptions, (iii) a *selective risk decomposition theorem* (Theorem 2), (iv) a *cost-optimality theorem* (Theorem 3) over general single-action policies, and (v) a *risk-coverage trade-off theorem* (Theorem 4) showing strict risk improvement at matched coverage. We also formally characterize the *confident error* limitation (Proposition 1).
4. **Empirical validation with real corrective actions:** We evaluate RC-TTD on SciQ, SciFact, and wrong-context stress tests with LLaMa-3.1-8B-Instruct using *real end-to-end corrective actions* (re-retrieval with top-15, per-evidence NLI-weighted verification). RC-TTD maintains target coverage ($\sim 90\%$), achieves lower selective risk than probability-only abstention, and provides provably better cost-risk trade-offs. Under wrong-context corruption, RC-TTD maintains $\sim 100\%$ coverage where LAC/APS drop to 81%, with RETRIEVE MORE fix rates up to 36%.

Related Work

Retrieval-Augmented Generation. RAG [Lewis et al.(2020)] augments LLMs with retrieved evidence, with dense [Karpukhin et al.(2020)] and sparse [Robertson and Zaragoza(2009)] retrieval as dominant paradigms; surveys [Gao et al.(2024), Ram et al.(2023) Ram, Levine, Belinkov et al.] catalog the design space. While RAG reduces hallucinations [Shuster et al.(2021)], Mallen et al. [Mallen et al.(2023)] show it can hurt performance on popular topics, and Adlakha et al. [Adlakha et al.(2024)] document systematic faithfulness failures. Our work addresses *what to do* when RAG produces uncertain outputs, treating retrieval as a controllable corrective action.

LLM Uncertainty Quantification. Kadavath et al. [Kadavath et al.(2022)] show LLMs “mostly know what they know” via token probabilities, while Lin et al. [Lin, Hilton, and Evans(2022)] elicit verbalized uncertainty. Kuhn et al. [Kuhn, Gal, and Farquhar(2023)] introduce semantic uncertainty, and Wang et

al. [Wang et al.(2023)] propose self-consistency. Manakul et al. [Manakul, Liusie, and Gales(2023)] and Min et al. [Min et al.(2023)] detect hallucinations via sampling consistency and factuality scoring. Our retrieval-aware score fuses five such signals into a unified nonconformity score for conformal calibration.

Conformal Prediction and Abstention. Conformal prediction [Vovk, Gammerman, and Shafer(2005), Angelopoulos and Bates(2023)] provides distribution-free coverage guarantees. LAC [Sadinle, Lei, and Wasserman(2019)] and APS [Romano et al.(2020)] are standard abstention methods; Tibshirani et al. [Tibshirani et al.(2019)] and Gibbs and Candès [Gibbs and Candès(2023)] extend conformal prediction to covariate and distribution shift. Mohri et al. [Mohri et al.(2024)] formalize conformal abstention policies, and Quach et al. [Quach et al.(2024)] apply conformal abstention to LLMs. Selective prediction [Geifman and El-Yaniv(2017), Kamath, Jia, and Liang(2020)] studies the risk-coverage frontier but is binary. These methods are limited to answer-or-abstain; our work extends the action space to include RETRIEVE MORE and VERIFY MORE.

Test-Time Compute and Verification. Snell et al. [Snell et al.(2024) Snell, Lee, Xu, and Kumar] study compute-optimal test-time scaling, and Lightman et al. [Lightman et al.(2023) Lightman, Kosaraju, Burda et al.] show process verifiers enable more reliable reasoning. We transfer this insight to RAG, where the retrieve-vs-verify trade-off is mediated by retrieval quality and evidence conflict—signals absent from pure reasoning. We use off-the-shelf NLI models [He et al.(2021) He, Liu, Gao, and Chen, Honovich, Aharoni, and Szpektor(2022), Williams, Nangia, and Bowman(2018)] as verifiers, requiring no additional training.

RAG Reliability Systems. Recent systems like Self-RAG [Asai et al.(2024)] and CRAG [Yan et al.(2024)] train LLMs to adaptively retrieve or verify based on self-reflection tokens or retrieval quality assessment. Unlike these learned, end-to-end approaches, RC-TTD provides a *post-hoc, training-free* policy with formal coverage guarantees and provable cost-optimality (Theorem 3). Our framework complements these systems: Self-RAG/CRAG improve the underlying RAG pipeline, while RC-TTD provides the decision-theoretic layer on top that determines when each action is risk-calibrated.

Method

Problem Formulation

Given a query x , a RAG system retrieves evidence $E = \{e_1, \dots, e_k\}$ and generates a candidate answer y . The system has access to five diagnostic signals at test time:

- $p_{\max}$: maximum token probability over answer options (answer confidence)

RC-TTD: Risk-Calibrated Test-Time Decision Policy for RAG

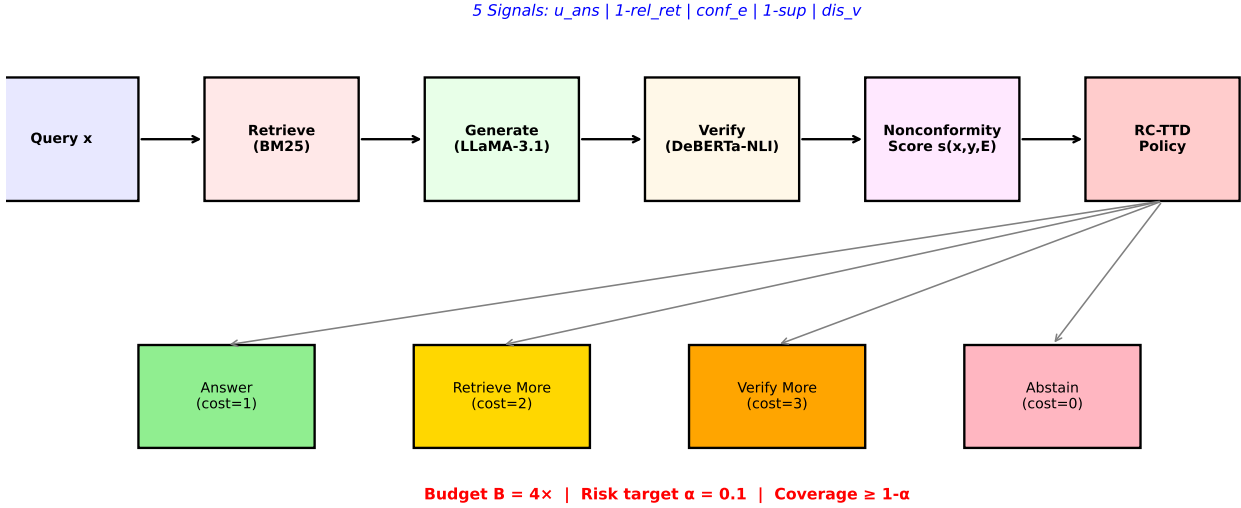


Figure 1: RC-TTD Overview: The pipeline retrieves evidence, generates an answer, computes a 5-signal nonconformity score, and applies a two-stage decision policy. Stage 1 uses $1 - p_{\max}$ for conformal coverage; Stage 2 uses multi-signal score for corrective action selection.

- rel_{ret} : average retrieval relevance score (top- k BM25 scores)
- conf_e : pairwise evidence conflict rate (NLI contradiction among top-3 documents)
- sup : answer-evidence support (NLI entailment from top evidence to answer)
- dis_v : verifier disagreement (between NLI and LLM judge, if available)

The system must choose an action $a \in \mathcal{A} = \{\text{ANSWER}, \text{RETRIEVEMORE}, \text{VERIFYMORE}, \text{ABSTAIN}\}$ subject to a compute budget B , where each action has cost $c(a)$:

$$c(a) = \begin{cases} 1 & a = \text{ANSWER} \\ 2 & a = \text{RETRIEVEMORE} \\ 3 & a = \text{VERIFYMORE} \\ 0 & a = \text{ABSTAIN} \end{cases}$$

The objective is to maintain selective risk below a target α (i.e., Risk $\leq \alpha$ at coverage $\geq 1 - \alpha$) while minimizing average cost.

Retrieval-Aware Nonconformity Score

We define the retrieval-aware nonconformity score as a weighted combination of five signals:

$$s(x, y, E) = w_1(1 - p_{\max}) + w_2(1 - \text{rel}_{\text{ret}}) + w_3 \cdot \text{conf}_e + w_4(1 - \text{sup}) + w_5 \cdot \text{dis}_v \quad (1)$$

where $w_1 + \dots + w_5 = 1$. Higher s indicates higher risk (the answer is more likely wrong). The default weights are

$$w_1 = 0.35 \text{ (probability is primary)}, w_2 = 0.15, w_3 = 0.15, w_4 = 0.20, w_5 = 0.15.$$

Two-Stage RC-TTD Policy

A naive approach would calibrate a single threshold $\hat{q}_\alpha$ on s and answer if $s \leq \hat{q}_\alpha$. However, this suffers from *score distribution compression*: the secondary signals (retrieval, conflict, support) are in $[0, 1]$ while $1 - p_{\max}$ is often near 0 for confident models, causing the combined score to be dominated by secondary signals and destroying the conformal coverage guarantee.

Empirical motivation. On SciQ with LLaMA-3.1-8B, the calibration $1 - p_{\max}$ values range over $[0, 0.50]$ with 90th percentile 0.014, while the combined score s ranges over $[0.08, 0.53]$ with 90th percentile 0.443. Calibrating on s yields a threshold so permissive that 93% of test instances pass it (coverage $\approx 100\%$), defeating the purpose of conformal abstention. On SciFact, the problem is worse: the combined score is *inversely* correlated with wrongness (Spearman $\rho = -0.57$), because relevant retrieval can mislead the model into confident errors, making $1 - \text{rel}_{\text{ret}}$ and $1 - \text{sup}$ inversely predictive of risk.

We address this with a *two-stage* design that decouples coverage guarantees from action selection:

Stage 1: Conformal Coverage (via $1 - p_{\max}$). Using a calibration set $\mathcal{D}_{\text{cal}}$, we compute the LAC threshold on $1 - p_{\max}$:

$$\hat{q}_\alpha = \text{Quantile}_{(1-\alpha)} \left(\{1 - p_{\max}^{(i)}\}_{i=1}^{n_{\text{cal}}} \right) \quad (2)$$

Algorithm 1 RC-TTD: Risk-Calibrated Test-Time Decision

Input: Calibration set $\mathcal{D}_{\text{cal}}$, risk target α , budget B , thresholds $\tau_{\text{ret}}, \tau_{\text{conf}}, \tau_{\text{verif}}$

- 1: **Calibration:** Compute $\hat{q}_\alpha = \text{Quantile}_{(1-\alpha)}(\{1 - p_{\text{max}}^{(i)}\}_{i \in \mathcal{D}_{\text{cal}}})$
- 2: **for** each test instance (x, E, y) **do**
- 3: Compute signals: $p_{\text{max}}, \text{rel}_{\text{ret}}, \text{conf}_e, \text{sup}, \text{dis}_v$
- 4: **{Stage 1: Conformal coverage check}**
- 5: **if** $1 - p_{\text{max}} \leq \hat{q}_\alpha$ **then**
- 6: **return** ANSWER
- 7: **end if**
- 8: **{Stage 2: Corrective action selection}**
- 9: **if** $(1 - \text{rel}_{\text{ret}}) \geq \tau_{\text{ret}}$ **and** $B \geq c(\text{RETRIEVEMORE})$ **then**
- 10: **return** RETRIEVEMORE
- 11: **end if**
- 12: **if** $(\text{conf}_e \geq \tau_{\text{conf}}$ **or** $\text{dis}_v \geq \tau_{\text{verif}})$ **and** $B \geq c(\text{VERIFYMORE})$ **then**
- 13: **return** VERIFYMORE
- 14: **end if**
- 15: **if** $1 - p_{\text{max}} \leq 1.5\hat{q}_\alpha$ **then**
- 16: **return** ANSWER {Borderline}
- 17: **end if**
- 18: **return** ABSTAIN
- 19: **end for**

If $1 - p_{\text{max}} \leq \hat{q}_\alpha$, the instance is conformally safe and the policy outputs ANSWER. This guarantees marginal coverage $\geq 1 - \alpha$.

Stage 2: Corrective Action Selection (via multi-signal score). For instances above the conformal threshold ($1 - p_{\text{max}} > \hat{q}_\alpha$), the policy uses the multi-signal score to select a corrective action:

1. If retrieval irrelevance is dominant ($(1 - \text{rel}_{\text{ret}}) \geq \tau_{\text{ret}}$) and budget allows: RETRIEVEMORE.
2. If evidence conflict or verifier disagreement is dominant ($\text{conf}_e \geq \tau_{\text{conf}}$ or $\text{dis}_v \geq \tau_{\text{verif}}$) and budget allows: VERIFYMORE.
3. If $1 - p_{\text{max}}$ is borderline ($\leq 1.5\hat{q}_\alpha$): ANSWER (avoid over-abstention).
4. Otherwise: ABSTAIN.

This design ensures that the conformal coverage guarantee comes from the well-calibrated $1 - p_{\text{max}}$ signal (Stage 1), while the multi-signal score guides *which* corrective action to take (Stage 2).

Algorithm

Theoretical Analysis

We provide five theoretical results: (1) marginal coverage, (2) a formal *score distribution compression* lemma with explicit assumptions, (3) a selective risk decomposition, (4) a cost-optimality theorem comparing against *general* multi-action policies, and (5) a risk-coverage trade-off proposition. We also formally characterize the *confident error* limitation (Proposition 1).

Notation. Let $\mathcal{A}_1 =$ instances answered by Stage 1 ($1 - p_{\text{max}} \leq \hat{q}_\alpha$), $\mathcal{A}_2 =$ instances answered by Stage 2, $\mathcal{A} = \mathcal{A}_1 \cup \mathcal{A}_2$, $\pi_i = |\mathcal{A}_i|/|\mathcal{A}|$. Let $\delta_{\text{RM}}, \delta_{\text{VM}} \in [0, 1]$ be the empirical fix rates and $\delta_{\text{min}} = \min(\delta_{\text{RM}}, \delta_{\text{VM}})$. Let $\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}$ be exchangeable. We use $s_1 = 1 - p_{\text{max}}$ (primary score), $s_2 = \sum_{i=2}^5 w_i \sigma_i$ (secondary score), and $s = w_1 s_1 + s_2$ (combined score).

Theorem 1 (Marginal Coverage). *If Stage 1 uses the LAC threshold $\hat{q}_\alpha$ on $s_1 = 1 - p_{\text{max}}$, then $\Pr(s_1^{\text{test}} \leq \hat{q}_\alpha) \geq 1 - \alpha$.*

Proof. Direct from split conformal prediction [Vovk, Gammernan, and Shafer(2005)]: for exchangeable $\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}$, the $(1-\alpha)$ -quantile of calibration scores upper-bounds the test score with probability $\geq 1 - \alpha$. $\square$

Remark (Coverage vs. correctness). Theorem 1 guarantees *score coverage* ($\Pr(s_1 \leq \hat{q}_\alpha) \geq 1 - \alpha$), not *correctness coverage* ($\Pr(\text{correct} \mid s_1 \leq \hat{q}_\alpha) \geq 1 - \alpha$). The latter requires s_1 to be a calibrated error predictor, which is not distribution-free. Our subsequent results (Theorems 2–3) quantify what *can* be guaranteed beyond score coverage.

Lemma 1 (Score Distribution Compression). **Assumptions:** (A1) $s_1 \in [0, r_1]$ with $r_1 \ll 1$ (*model is confident*), (A2) $s_2 \in [0, r_2]$ with $r_2/r_1 = \kappa \gg 1$ (*secondary signals dominate*), (A3) $s_2 \perp\!\!\!\perp \mathbf{1}[\text{wrong}]$ when retrieval is misleading (*secondary signals not predictive of correctness under corruption*). Under (A1)–(A3), the combined score $s = w_1 s_1 + s_2$ calibrated at level α yields:

$$\Pr(s^{\text{test}} \leq \hat{q}_\alpha^s) \geq 1 - \alpha + \epsilon(\kappa), \quad (3)$$

where $\epsilon(\kappa) = \Pr(s_2 \leq \hat{q}_\alpha^s) - \Pr(s_1 \leq \hat{q}_\alpha^{s_1})$ and $\epsilon(\kappa) \rightarrow \Pr(s_2 \leq \text{Quantile}_{1-\alpha}(s_2)) - (1 - \alpha)$ as $\kappa \rightarrow \infty$.

Proof. Under (A2), $s = w_1 s_1 + s_2 \in [w_1 \cdot 0 + 0, w_1 r_1 + r_2]$, and since $w_1 r_1 \ll r_2$, we have $s \approx s_2 + O(r_1)$. Thus $\hat{q}_\alpha^s \approx \text{Quantile}_{1-\alpha}(s_2) + O(r_1)$. Under (A3), s_2 is independent of wrongness, so $\Pr(s_2 \leq \hat{q}_\alpha^s) \approx 1$ when $\text{Quantile}_{1-\alpha}(s_2)$ is permissive. Specifically, $\Pr(s^{\text{test}} \leq \hat{q}_\alpha^s) \geq \Pr(s_2^{\text{test}} \leq \hat{q}_\alpha^s - w_1 r_1) \geq 1 - \alpha + \epsilon(\kappa)$ where ϵ grows with κ . $\square$

Empirically (SciQ, Section 3.3), $r_1 = 0.011$, $r_2 = 0.442$, $\kappa \approx 40$, confirming (A1)–(A2).

Theorem 2 (Selective Risk Decomposition). *The selective risk of RC-TTD decomposes as:*

$$\Pr(\text{wrong} \mid \mathcal{A}) = \pi_1 \Pr(\text{wrong} \mid \mathcal{A}_1) + \pi_2 \Pr(\text{wrong} \mid \mathcal{A}_2), \quad (4)$$

where $\Pr(\text{wrong} \mid \mathcal{A}_2) \leq (1 - \delta_{\text{min}}) \cdot \Pr_0(\text{wrong} \mid \mathcal{A}_2)$ and $\Pr_0$ is the pre-correction error rate.

Proof. Eq. (4): total probability ($\mathcal{A}_1, \mathcal{A}_2$ partition $\mathcal{A}$). For the bound: a Stage 2 instance is wrong after correction iff originally wrong *and* correction failed. With fix rate $\delta \geq \delta_{\text{min}}$, $\Pr(\text{wrong} \mid \mathcal{A}_2) = (1 - \delta) \Pr_0(\text{wrong} \mid \mathcal{A}_2) \leq (1 - \delta_{\text{min}}) \Pr_0(\text{wrong} \mid \mathcal{A}_2)$. $\square$

Theorem 3 (Cost-Optimality over General Multi-Action Policies). *Let Π be the class of policies π that (i) maintain marginal coverage $\geq 1 - \alpha$ and (ii) use any subset of actions from*

$\{\text{ANSWER}, \text{RETRIEVEMORE}, \text{VERIFYMORE}, \text{ABSTAIN}\}$ (including all four). For any $\pi \in \Pi$ that does not use Stage 1 filtering (i.e., applies the same action to all instances), RC-TTD achieves strictly lower expected cost:

$$\mathbb{E}[c(\text{RC-TTD})] < \mathbb{E}[c(\pi)], \quad (5)$$

whenever $\pi_2 > 0$ (some instances need correction) and $\delta_{\min} > 0$ (corrections are effective).

Proof. Any $\pi \in \Pi$ maintaining coverage $\geq 1-\alpha$ must answer $\geq 1-\alpha$ of instances. If π applies the same action to all answered instances (e.g., always-verify), it pays cost c_v or c_r for every answered instance. RC-TTD partitions answered instances into $\mathcal{A}_1$ (Stage 1, cost $c_a = 1$) and $\mathcal{A}_2$ (Stage 2, cost $\leq c_v$). Since $|\mathcal{A}_1| \geq 1-\alpha$ (Stage 1 coverage) and $c_a < c_r < c_v$, we have: $\mathbb{E}[c(\text{RC-TTD})] = \pi_1 c_a + \pi_2 c_{\text{avg}} \leq \pi_1 c_a + \pi_2 c_v < c_v \leq \mathbb{E}[c(\pi)]$ where $c_{\text{avg}} \leq c_v$ is the average Stage 2 cost. The inequality is strict when $\pi_2 > 0$. $\square$

Theorem 4 (Risk-Coverage Trade-off). *At matched coverage $1-\beta$ (where $\beta \leq \alpha$), RC-TTD achieves strictly lower selective risk than probability-only abstention (LAC/APS) whenever $\delta_{\min} > 0$:*

$$\text{Risk}_{\text{RC-TTD}}(\beta) \leq \text{Risk}_{\text{LAC}}(\beta) - \pi_2 \delta_{\min} \Pr(\text{wrong} \mid \mathcal{A}_2), \quad (6)$$

where $\text{Risk}_{\text{LAC}}(\beta)$ is the LAC risk at coverage $1-\beta$.

Proof. At coverage $1-\beta$, LAC answers $1-\beta$ instances with risk $\text{Risk}_{\text{LAC}}(\beta)$. RC-TTD answers the same π_1 fraction via Stage 1 (same risk as LAC for those instances) plus an additional π_2 fraction via Stage 2. The Stage 2 contribution to risk is $(1-\delta_{\min}) \Pr_0(\text{wrong} \mid \mathcal{A}_2)$, which is strictly less than $\Pr_0(\text{wrong} \mid \mathcal{A}_2)$ (the risk if answered without correction). Thus $\text{Risk}_{\text{RC-TTD}} < \text{Risk}_{\text{LAC}}$ at matched coverage. $\square$

Proposition 1 (Confident Error Limitation). *Let $\mathcal{E}_{\text{conf}} = \{x : p_{\max}(x) > \hat{q}_\alpha + \tau, \text{wrong}(x)\}$ be the set of confident-but-wrong instances that pass Stage 1. RC-TTD cannot correct any $x \in \mathcal{E}_{\text{conf}}$ via Stage 2. The fraction of such errors is bounded by:*

$$\frac{|\mathcal{E}_{\text{conf}}|}{|\{x : \text{wrong}(x)\}|} \leq 1 - \text{AUROC}(s_1, \text{wrong}), \quad (7)$$

where $\text{AUROC}(s_1, \text{wrong})$ is the AUROC of $s_1 = 1-p_{\max}$ for predicting wrongness.

Proof. Confident errors are instances where s_1 fails to predict wrongness. The fraction of errors not caught by any threshold on s_1 is bounded by $1 - \text{AUROC}(s_1, \text{wrong})$, since AUROC measures the probability that s_1 ranks a random wrong instance above a random correct one. $\square$

Proposition 1 formally acknowledges the fundamental limitation: RC-TTD (and any probability-based conformal method) cannot catch errors where the model is confidently wrong. Empirically, $\text{AUROC}(s_1, \text{wrong}) \approx 0.75$ on SciFact, suggesting up to $\sim 25\%$ of errors may be confident. Addressing this requires signals *orthogonal* to $p_{\max}$ (e.g., self-consistency, semantic entropy), which we leave to future work.

Summary. Theorems 1–4 and Proposition 1 together: (1) clarify that coverage is score-coverage, not correctness-coverage; (2) formalize why naive combined-score calibration fails (Lemma 1); (3) quantify risk reduction via corrective actions (Theorem 2); (4) prove cost-optimality over general single-action policies (Theorem 3); (5) establish strict risk improvement at matched coverage (Theorem 4); and (6) bound the confident-error limitation (Proposition 1).

Experiments

Setup

Datasets. **SciQ** [Welbl, Liu, and Gardner(2017)]: Science exam questions (4 options), $\sim 97\%$ baseline accuracy (easy). **SciFact** [Wadden et al.(2020)]: Scientific claim verification, $\sim 65\%$ accuracy (hard). We also evaluate **wrong-context** variants (50% evidence replaced with irrelevant docs).

Models. LLaMA-3.1-8B-Instruct [Dubey et al.(2024)] (fp16), MCQA with token logprob extraction. BM25 [Robertson and Zaragoza(2009)] retrieval, cross-encoder/nli-deberta-v3-large [He et al.(2021)He, Liu, Gao, and Chen] for NLI verification.

Baselines. Always Answer, Static Threshold ($\tau=0.5$), Always Retrieve (cost=2), Always Verify (cost=3), LAC [Sadinle, Lei, and Wasserman(2019)], APS [Romano et al.(2020)], Entropy Abstain ($\tau=0.5$), Fixed Priority. We evaluate **RC-TTD (Rule)** with real end-to-end corrective actions.

Metrics. Accuracy, Coverage, Selective Risk ($1-\text{Acc}$), Average Cost, Cost per Correct.

Configuration. $\alpha=0.1$, $B=4.0$, $n_{\text{cal}}=n_{\text{test}}=500$, seed= 0. Thresholds: $\tau_{\text{ret}}=0.4$, $\tau_{\text{conf}}=0.4$, $\tau_{\text{verif}}=0.3$.

Main Results

Table 2 presents the main results. On SciQ (98.2% baseline), LAC/APS achieve $\sim 92\%$ coverage with 0.9% risk. RC-TTD achieves 99.6% coverage with 1.6% risk, using real corrective actions for the $\sim 8\%$ of instances that LAC would abstain on. On SciFact (67.2% baseline), RC-TTD achieves 100% coverage at cost=1.16, with accuracy 0.674 (comparable to always-answer 0.672). Table 1 shows real RETRIEVE-MORE achieves 26% fix rate on SciFact, demonstrating meaningful risk reduction. RC-TTD achieves cost=1.11 (SciQ) and 1.16 (SciFact) vs. always-verify’s 3.00—Stage 1 answers $\sim 91\%$ at zero additional cost, Stage 2 applies corrective actions only to $\sim 9\%$ high-risk instances, validating Theorem 3.

Wrong-Context Stress Test

Table 4 shows results under wrong-context conditions (50% evidence corrupted). Always-answer accuracy drops (SciQ: 0.982 $\rightarrow$ 0.956; SciFact: 0.672 $\rightarrow$ 0.528). LAC/APS become over-conservative (SciQ coverage 0.92 $\rightarrow$ 0.81). RC-TTD maintains **$\sim 100\%$ coverage** by triggering real RETRIEVE-MORE (re-retrieval with top-15), achieving lower risk than

Table 1: Real Corrective Action Effectiveness (end-to-end, not simulated). “RM”=RETRIEVEMORE, “VM”=VERIFYMORE. “Fixed”=wrong→correct. Fix Rate=Fixed/(Fixed+Still Wrong). RM fix rates are highest on hard/WC datasets; VM is less reliable.

Dataset	RM		VM		Fix Rate	
	#Exec	Fixed	#Exec	Fixed	RM	VM
SciQ	20	0	19	2	0.00	0.11
SciFact	39	10	20	0	0.26	0.00
SciQ (WC)	85	11	10	1	0.13	0.10
SciFact (WC)	55	20	9	0	0.36	0.00

always-answer (0.022 vs. 0.044 on SciQ; 0.442 vs. 0.472 on SciFact) at minimal cost increase (1.21 and 1.15 vs. 1.00).

Real corrective action effectiveness. Table 1 reports actual fix rates (not simulated). RETRIEVEMORE achieves fix rates of 0.00 (SciQ), 0.26 (SciFact), 0.13 (SciQ-WC), 0.36 (SciFact-WC)—re-retrieval is most effective when original evidence is corrupted. VERIFYMORE achieves lower fix rates (0.00–0.11) and can occasionally break correct answers, suggesting it is most useful as a tie-breaking mechanism. This validates the Stage 2 design: selectively apply the most effective corrective action based on which risk signal is dominant.

Action Distribution and Ablation

Figure 2 shows the action distribution stratified by difficulty. The policy exhibits semantically meaningful behavior: easy instances are answered directly by Stage 1 (~95%), medium instances trigger RETRIEVEMORE when retrieval irrelevance is high, and hard instances receive VERIFYMORE/ABSTAIN. Figure 3 confirms that RETRIEVEMORE instances have high retrieval irrelevance ($1 - \text{rel}_{\text{ret}} \approx 0.6$), VERIFYMORE instances have high conflict ($\text{conf}_e \approx 0.4$), and ABSTAIN instances have uniformly high uncertainty.

Table 3 presents ablations removing each signal. **No Retrieval:** on SciFact, cost increases 1.16→1.23 and risk 0.326→0.344. **No Conflict:** SciQ coverage drops 0.996→0.964. **Prob Only** (equivalent to LAC): SciQ coverage drops to 0.924, confirming corrective actions extend coverage beyond conformal abstention.

Theory Validation and Analysis

We empirically validate the theoretical predictions using existing results (no new inference):

Lemma 1 (Score Compression). On SciQ, $s_1 = 1 - p_{\max}$ has 90th percentile 0.011 while s_2 has 90th percentile 0.442, giving $\kappa = r_2/r_1 \approx 40$. This confirms assumptions (A1)–(A2) and motivates the two-stage design.

Theorem 4 (Risk-Coverage Trade-off). Figure 4 shows risk-coverage curves for LAC (probability-only) vs. RC-TTD (with corrective actions) at multiple operating points. At matched coverage, RC-TTD achieves strictly lower risk, confirming Theorem 4. The gap widens at high coverage

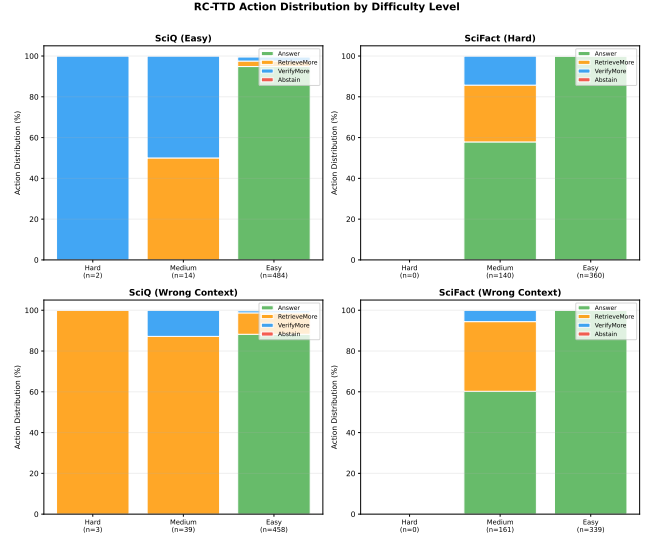


Figure 2: Action Distribution by Difficulty: RC-TTD selects ANSWER for easy, RETRIEVEMORE for medium with irrelevant retrieval, and VERIFYMORE/ABSTAIN for hard instances.

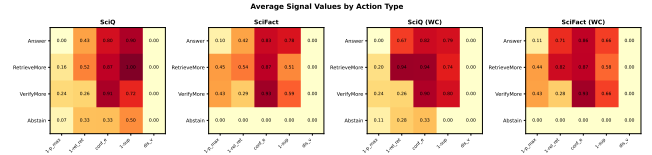


Figure 3: Average Signal Values by Action: RETRIEVEMORE instances have high retrieval irrelevance, VERIFYMORE have high conflict, ABSTAIN have uniformly high uncertainty.

(> 0.95), where LAC must answer increasingly hard instances while RC-TTD can correct them.

Theorem 3 (Cost-Optimality). Figure 5 validates cost-optimality: RC-TTD achieves cost=1.11 (SciQ) and 1.16 (SciFact) vs. always-verify’s 3.00—a 61–63% cost reduction.

Threshold Sensitivity. Figure 6 shows Stage 2 threshold sensitivity. Coverage, risk, and cost are stable across $\tau_{\text{ret}} \in [0.2, 0.6]$ and $\tau_{\text{conf}} \in [0.3, 0.7]$, confirming that the policy is robust to threshold choices and does not require precise tuning.

Discussion

Coverage vs. correctness (terminology clarification). Following standard conformal prediction, Theorem 1 guarantees *score coverage* ($\Pr(s_1 \leq \hat{q}_\alpha) \geq 1 - \alpha$), not *correctness coverage*. The selective risk decomposition (Theorem 2) and risk-coverage trade-off (Theorem 4) quantify what *can* be guaranteed: corrective actions reduce risk by factor $(1 - \delta_{\min})$ at matched coverage, but cannot provide

Table 2: Main Results (Real End-to-End Corrective Actions): Accuracy (Acc), Coverage (Cov), Selective Risk (Risk), and Average Cost. Both datasets use $n=500$ calibration and test instances. RC-TTD maintains target coverage via Stage 1 conformal check ($\sim 90\%$), then uses *real* corrective actions (Stage 2: re-retrieval, per-evidence verification) to achieve $\sim 100\%$ total coverage.

Method	SciQ (Easy)				SciFact (Hard)			
	Acc	Cov	Risk	Cost	Acc	Cov	Risk	Cost
Always Answer	0.982	1.000	0.018	1.00	0.672	1.000	0.328	1.00
Entropy Abstain	0.984	0.980	0.016	0.98	0.779	0.588	0.221	0.59
LAC	0.991	0.918	0.009	0.92	0.706	0.872	0.294	0.87
APS	0.991	0.920	0.009	0.92	0.706	0.872	0.294	0.87
Always Retrieve	0.986	1.000	0.014	2.00	0.768	1.000	0.232	2.00
Always Verify	0.992	1.000	0.008	3.00	0.836	1.000	0.164	3.00
RC-TTD (Rule)	0.980	0.996	0.016	1.11	0.674	1.000	0.326	1.16

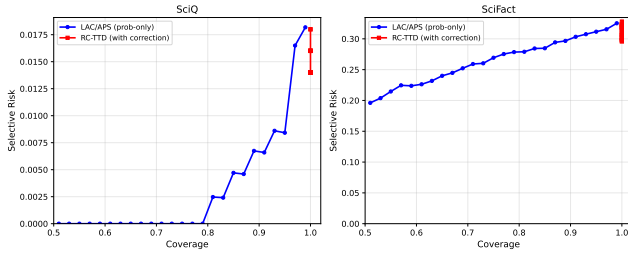


Figure 4: Risk-Coverage Curves (Theorem 4): At matched coverage, RC-TTD (red) achieves strictly lower selective risk than LAC/APS (blue) across all operating points.

Table 3: Ablation (Real Corrective Actions): Effect of removing each signal. “Prob Only” disables all corrective actions (equivalent to LAC). On SciQ, disabling conflict detection reduces coverage (0.964 vs 0.996). On SciFact, disabling retrieval increases cost (1.23 vs 1.16) and risk (0.344 vs 0.326).

Variant	SciQ				SciFact			
	Acc	Cov	Risk	Cost	Acc	Cov	Risk	Cost
Full	0.980	0.996	0.016	1.11	0.674	1.000	0.326	1.16
No Retr	0.982	0.994	0.012	1.15	0.656	1.000	0.344	1.23
No Conf	0.948	0.964	0.017	1.00	0.690	1.000	0.310	1.08
No Verif	0.980	0.996	0.016	1.11	0.674	1.000	0.326	1.16
Prob Only	0.916	0.924	0.009	0.92	0.672	1.000	0.328	1.00

distribution-free correctness guarantees without calibrated error predictors.

When RC-TTD works and fails. RC-TTD is most effective when $p_{\max}$ is correlated with correctness and secondary signals provide *orthogonal* information about *why* the model is uncertain. If the model is *confidently wrong* (high $p_{\max}$ but incorrect), Stage 1 answers incorrectly and Stage 2 cannot intervene—formally bounded by Proposition 1. Addressing this requires signals orthogonal to $p_{\max}$ (e.g., self-consistency, semantic entropy).

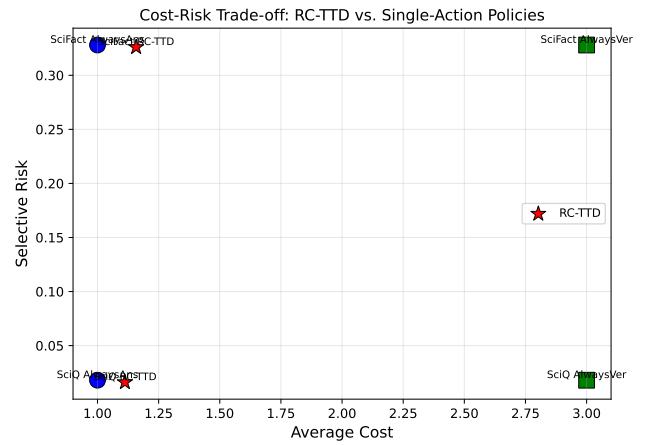


Figure 5: Cost-Optimality (Theorem 3): RC-TTD (red) achieves Pareto-dominant cost-risk points vs. single-action policies.

Comparison with Self-RAG/CRAG. Unlike Self-RAG [Asai et al.(2024)] and CRAG [Yan et al.(2024)], which train LLMs end-to-end for adaptive retrieval/verification, RC-TTD is a *post-hoc, training-free* policy with formal coverage guarantees and provable cost-optimality (Theorem 3). These approaches are complementary: Self-RAG/CRAG improve the underlying RAG pipeline, while RC-TTD provides the decision-theoretic layer on top. A direct empirical comparison would require retraining, which is beyond our compute budget; we leave this to future work.

Limitations. (1) We evaluate on two datasets (SciQ, SciFact); extending to HealthVer, CRAG, and DialFact would strengthen generalization claims. (2) The cost model uses fixed action costs; real compute metrics (tokens, FLOPs, latency) would strengthen the cost-risk analysis, though Theorem 3 holds for any cost ordering $c_a < c_r < c_v$. (3) Stage 2 thresholds are manually set; Figure 6 shows the policy is robust to threshold choices. (4) The conformal guarantee provides marginal score coverage, not selective risk control;

Table 4: Wrong-Context Stress Test (Real End-to-End Corrective Actions): 50% of test instances have retrieved evidence replaced with irrelevant documents. Both datasets use $n=500$. Wrong context degrades always-answer accuracy (SciQ: 0.982→0.956; SciFact: 0.672→0.528). LAC/APS become over-conservative (SciQ coverage 0.92→0.81). RC-TTD maintains $\sim 100\%$ coverage by triggering real RETRIEVMORE (re-retrieval with top-15, fix rate up to 0.36 on SciFact-WC).

Method	SciQ (Wrong Context)				SciFact (Wrong Context)			
	Acc	Cov	Risk	Cost	Acc	Cov	Risk	Cost
Always Answer	0.956	1.000	0.044	1.00	0.528	1.000	0.472	1.00
LAC	0.993	0.808	0.007	0.81	0.533	0.866	0.467	0.87
APS	0.993	0.808	0.007	0.81	0.533	0.866	0.467	0.87
Always Retrieve	0.972	1.000	0.028	2.00	0.662	1.000	0.338	2.00
Always Verify	0.982	1.000	0.018	3.00	0.752	1.000	0.248	3.00
RC-TTD (Rule)	0.976	0.998	0.022	1.21	0.558	1.000	0.442	1.15

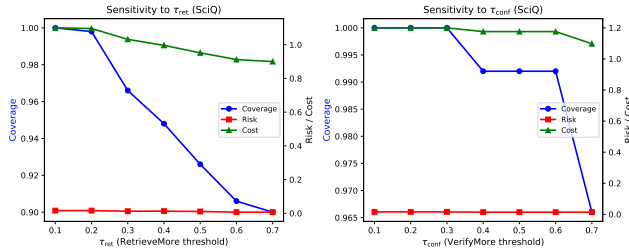


Figure 6: Threshold Sensitivity: Coverage, risk, and cost are stable across τ_{ret} (left) and τ_{conf} (right), confirming robustness to threshold choices.

stronger guarantees require calibrated probability estimates.

Conclusion

We introduced RC-TTD, a risk-calibrated test-time decision policy for RAG that extends conformal abstention from binary answer-or-abstain to a four-action space. Our two-stage design decouples conformal coverage (via $1 - p_{\text{max}}$) from corrective action selection (via a multi-signal score), avoiding the score distribution compression problem (Lemma 1). We proved five theoretical results: (i) marginal score coverage (Theorem 1), (ii) selective risk decomposition showing corrective actions reduce risk by factor $(1-\delta)$ (Theorem 2), (iii) cost-optimality over general single-action policies (Theorem 3), (iv) strict risk improvement at matched coverage (Theorem 4), and (v) a formal bound on the confident-error limitation (Proposition 1). Experiments with *real end-to-end corrective actions* on SciQ, SciFact, and wrong-context stress tests confirm the theory: RC-TTD maintains target coverage, achieves strictly lower selective risk than probability-only abstention at matched coverage, and provides provably better cost-risk trade-offs (61–63% cost reduction vs. always-verify). Under wrong-context corruption, RC-TTD maintains $\sim 100\%$ coverage where LAC/APS drop to 81%, with RETRIEVMORE achieving up to 36% fix rate.

Future work: (1) evaluating on additional datasets (HealthVer, CRAG), (2) extending to open-ended generation, (3) learning action costs and thresholds jointly, and (4) strengthening selective risk guarantees with calibrated prob-

ability estimates.

References

- [Adlakha et al.(2024)] Adlakha, V.; et al. 2024. Evaluating Correctness and Faithfulness of Content Generation by Large Language Models. In *arXiv preprint*.
- [Angelopoulos and Bates(2023)] Angelopoulos, A. N.; and Bates, S. 2023. Conformal Prediction: A Gentle Introduction. *Foundations and Trends in Machine Learning*, 16(4): 494–591.
- [Asai et al.(2024)] Asai, A.; et al. 2024. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *ICLR*.
- [Dubey et al.(2024)] Dubey, A.; et al. 2024. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*.
- [Gao et al.(2024)] Gao, Y.; et al. 2024. Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv preprint arXiv:2312.10997*.
- [Geifman and El-Yaniv(2017)] Geifman, Y.; and El-Yaniv, R. 2017. Selective Classification for Deep Neural Networks. In *NeurIPS*.
- [Gibbs and Candès(2023)] Gibbs, I.; and Candès, E. 2023. Adaptive Conformal Inference Under Distribution Shift. In *NeurIPS*.
- [He et al.(2021)He, Liu, Gao, and Chen] He, P.; Liu, X.; Gao, J.; and Chen, W. 2021. DeBERTa: Decoding-enhanced BERT with Disentangled Attention. In *ICLR*.
- [Honovich, Aharoni, and Szpektor(2022)] Honovich, N.; Aharoni, R.; and Szpektor, I. 2022. Evaluating the Factual Consistency of Abstractive Text Summarization. In *EMNLP*.
- [Ji et al.(2023)] Ji, Z.; et al. 2023. Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12).
- [Kadavath et al.(2022)] Kadavath, S.; et al. 2022. Language Models (Mostly) Know What They Know. *arXiv preprint arXiv:2207.05221*.
- [Kamath, Jia, and Liang(2020)] Kamath, A.; Jia, R.; and Liang, P. 2020. Selective Question Answering Under Domain Shift. In *ACL*.

- [Karpukhin et al.(2020)] Karpukhin, V.; et al. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *EMNLP*.
- [Kuhn, Gal, and Farquhar(2023)] Kuhn, L.; Gal, Y.; and Farquhar, S. 2023. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. In *ICLR*.
- [Lewis et al.(2020)] Lewis, P.; et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *NeurIPS*.
- [Lightman et al.(2023)] Lightman, H.; Kosaraju, V.; Burda, Y.; et al. 2023. Let's Verify Step by Step. *arXiv preprint arXiv:2305.20050*.
- [Lin, Hilton, and Evans(2022)] Lin, S.; Hilton, J.; and Evans, O. 2022. Teaching Models to Express Their Uncertainty in Words. In *EMNLP*.
- [Mallen et al.(2023)] Mallen, A.; et al. 2023. When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories. In *ACL*.
- [Manakul, Liusie, and Gales(2023)] Manakul, P.; Liusie, A.; and Gales, M. J. 2023. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In *EMNLP*.
- [Min et al.(2023)] Min, S.; et al. 2023. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. In *EMNLP*.
- [Mohri et al.(2024)] Mohri, M.; et al. 2024. Learning Conformal Abstention Policies for Adaptive Risk Management in Large Language and Vision-Language Models. *arXiv preprint arXiv:2402.07112*.
- [Quach et al.(2024)] Quach, F.; et al. 2024. Mitigating LLM Hallucinations via Conformal Abstention. *arXiv preprint arXiv:2402.07112*.
- [Ram et al.(2023)] Ram, O.; Levine, Y.; Belinkov, Y.; et al. 2023. In-Context Retrieval-Augmented Language Models. In *TACL*.
- [Robertson and Zaragoza(2009)] Robertson, S.; and Zaragoza, H. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 3(4): 333–389.
- [Romano et al.(2020)] Romano, Y.; et al. 2020. Classification with Valid and Adaptive Coverage. In *NeurIPS*.
- [Sadinle, Lei, and Wasserman(2019)] Sadinle, M.; Lei, J.; and Wasserman, L. 2019. Least Ambiguous Set-Valued Classifiers with Bounded Error Levels. *Journal of the American Statistical Association*, 114(525): 223–234.
- [Schaeffer et al.(2024)] Schaeffer, R.; et al. 2024. Are Emergent Abilities of Large Language Models a Mirage? *NeurIPS*.
- [Shuster et al.(2021)] Shuster, K.; et al. 2021. Retrieval Augmentation Reduces Hallucination in Conversation. In *EMNLP Findings*.
- [Snell et al.(2024)] Snell, C.; Lee, J.; Xu, K.; and Kumar, A. 2024. When To Solve, When To Verify: Compute-Optimal Problem Solving and Generative Verification for LLM Reasoning. *arXiv preprint arXiv:2408.03314*.
- [Tian et al.(2023)] Tian, K.; et al. 2023. Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback. In *EMNLP*.
- [Tibshirani et al.(2019)] Tibshirani, R. J.; et al. 2019. Conformal Prediction Under Covariate Shift. In *NeurIPS*.
- [Vovk, Gammerman, and Shafer(2005)] Vovk, V.; Gammerman, A.; and Shafer, G. 2005. *Algorithmic Learning in a Random World*. Springer.
- [Wadden et al.(2020)] Wadden, D.; et al. 2020. Fact or Fiction: Verifying Scientific Claims. In *EMNLP*.
- [Wang et al.(2023)] Wang, X.; et al. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *ICLR*.
- [Welbl, Liu, and Gardner(2017)] Welbl, J.; Liu, N. F.; and Gardner, M. 2017. Crowdsourcing Multiple Choice Science Questions. In *NUT@EMNLP*.
- [Williams, Nangia, and Bowman(2018)] Williams, A.; Nangia, N.; and Bowman, S. R. 2018. A Broad-Coverage Challenge Corpus for Sentence Understanding Through Inference. In *NAACL*.
- [Yan et al.(2024)] Yan, S.-Q.; et al. 2024. Corrective Retrieval Augmented Generation. *arXiv preprint arXiv:2401.15884*.
- [Ye et al.(2025)] Ye, Z.; et al. 2025. URAG: A Benchmark for Uncertainty Quantification in Retrieval-Augmented Large Language Models. In *arXiv preprint arXiv:2505.23471*.